Appendix

Appendix A
Plots

Figure 1, 2, 3, and 4 present the PI plots generated by the best-performing model under the ensembled setting.

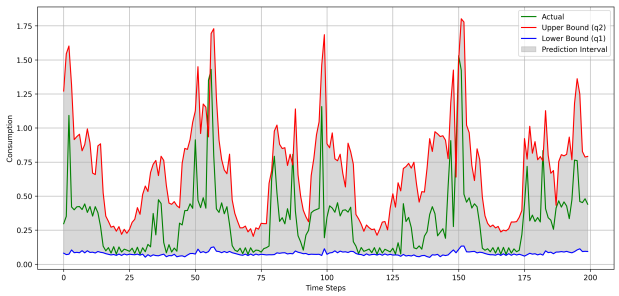

Fig. 1: PI obtained by the best performing model: ensembled TCN Tube for Dataset A.

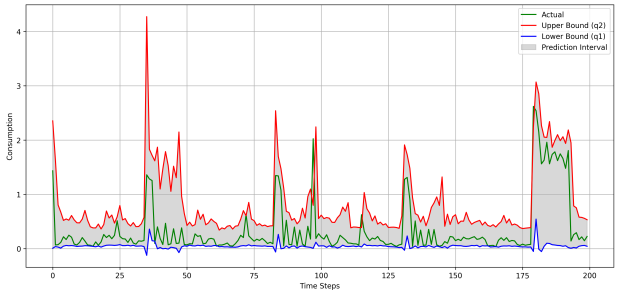

Fig. 2: PI obtained by the best performing model: ensembled TCN Tube for Dataset B.

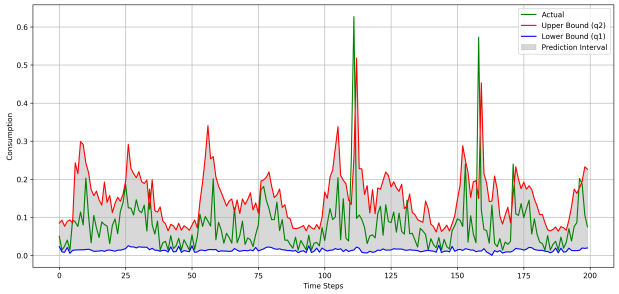

Fig. 3: PI obtained by the best performing model: ensembled TCN Tube for Dataset C

Appendix B
Model Hyperparameters

For the numerical experiments conducted on the LCL dataset, we considered majorly four modeLs: Tube-based

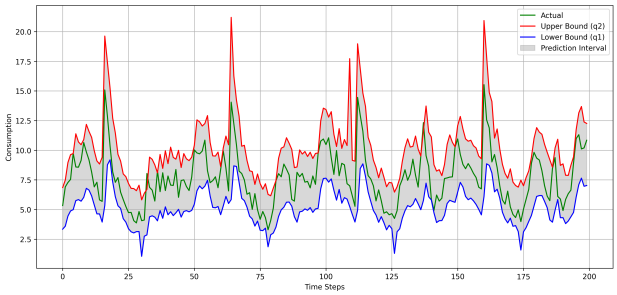

Fig. 4: PI obtained by the best performing model: ensembled TCN Tube for Dataset D

neural networks, quantile-based neural networks, DeepAR, and Mixture Density Networks (MDNs). For each dataset (Dataset A, Dataset B, Dataset C, and Dataset D), we tune the hyperparameters separately and select the configurations that provide the best balance between predictive coverage and interval sharpness on the test set. In this section, we report the tuning details for Dataset A. Tables for the other datasets (Dataset B, Dataset C, and Dataset D) are provided in subsequent subsections.

Hyperparameter Tuning Strategy

Across all datasets, we follow a consistent tuning protocol to ensure fair comparisons across model families. Choices of hyperparameters were made through validation experiments, balancing interval coverage (PICP), width (MPIW), and training stability.

a) Tube models: For Tube Loss-based models (LSTM-Tube, TCN-Tube, GRU-Tube), we tuned $r \in \{0.1, 0.2, \ldots, 0.5\}$ and $\delta \in \{0.001, 0.005, 0.01, 0.1\}$. Tuning follows a two-stage process: (i) fix $\delta = 0$ and sweep r to match the target coverage $t = 0.95$; (ii) if coverage exceeds t, increase δ gradually to reduce tube width while maintaining coverage. Window sizes are searched from $\{6, 12, 24, 48\}$ and batch sizes from $\{16, 32, 64\}$. The learning rate is set to 10^{-3} with Adam and decay scheduling.

b) Quantile models: For quantile-based models (LSTM, TCN, GRU), prediction intervals were always constructed with fixed quantiles $q_1 = 0.025$ and $q_2 = 0.975$, targeting 95% coverage. Unlike the quantiles, window size and batch size were not fixed globally; they were selected per dataset from $\{6, 12, 24, 48\}$ and $\{16, 32, 64\}$ respectively through validation. The learning rate was tuned from $\{0.1, 0.01, 0.001\}$, with 0.01 typically providing the best stability. All models were trained for 100 epochs using Adam.

c) DeepAR: For DeepAR, we explored the same grids of window size $\{6, 12, 24, 48\}$ and batch size $\{16, 32, 64\}$. The

final configuration usually included two recurrent layers with 64 units each, followed by a dense ReLU layer of 32 units and a Gaussian likelihood output. A learning rate of 10^{-3} yielded the most consistent validation log-likelihood across datasets.

d) MDN: For the Mixture Density Network (MDN), we tuned the number of mixture components and learning rate using standard grids, selecting the configuration that achieved the best validation log-likelihood with sharp and well-calibrated intervals. (Per-dataset settings are reported in the appendix further.)

TABLE I: Hyperparameters for Dataset A across models

Models	Hyperparameters							
Tube Models	Window Size	Batch Size	Structure	r	δ	q	learning rate	epochs
LSTM Tube	48	64	[64(0.3), 32(0.3)]	0.5	0.005	0.95	0.01	100
TCN Tube	24	64	[64(0.3)] × dilations(1,2,4,8)	0.7	0.002	0.95	0.001	100
GRU Tube	6	64	[64(0.3), 32(0.3)]	0.5	0	0.95	0.01	100
Quantile Models	Window Size	Batch Size	Structure	q1	q2	learning rate	epochs	
LSTM Quantile	48	64	[64(0), 32(0)]	0.025	0.975	0.01	100	
TCN Quantile	48	64	[64(0.3)] × dilations(1,2,4,8)	0.025	0.975	0.01	100	
GRU Quantile	48	64	[64(0.3), 32(0.3)]	0.025	0.975	0.01	100	
DeepAR Model	Window Size	Batch Size	Structure	learning rate				
DeepAR	24	16	[64 × 2, 32(ReLU), Gaussian]	1e-3				
MDN Model	Window Size	Batch Size	Structure	learning rate	Components			
MDN	6	16	[64(0)]	0.001	5			

TABLE II: Hyperparameters for Dataset B across models

Models	Hyperparameters							
Tube Models	Window Size	Batch Size	Structure	r	δ	q	learning rate	epochs
LSTM Tube	48	64	[64(0.3), 32(0.3)]	0.5	0	0.95	0.01	100
TCN Tube	48	64	[64(0.3)] × dilations(1,2,4,8)	0.4	0.005	0.95	0.001	100
GRU Tube	12	64	[64(0.3), 32(0.3)]	0.5	0.015	0.95	0.01	100
Quantile Models	Window Size	Batch Size	Structure	q1	q2	learning rate	epochs	
LSTM Quantile	48	64	[64(0), 32(0)]	0.025	0.975	0.01	100	
TCN Quantile	24	64	[64(0.3)] × dilations(1,2,4,8)	0.025	0.975	0.01	100	
GRU Quantile	48	64	[64(0.3), 32(0.3)]	0.025	0.975	0.01	100	
DeepAR Model	Window Size	Batch Size	Structure	learning rate				
DeepAR	48	32	[64 × 2, 32(ReLU), Gaussian]	1e-3				
MDN Model	Window Size	Batch Size	Structure	learning rate	Components			
MDN	48	64	[64(0)]	0.001	5			

TABLE III: Hyperparameters for Dataset C across models

Models	Hyperparameters							
Tube Models	Window Size	Batch Size	Structure	r	δ	q	learning rate	epochs
LSTM Tube	6	64	[64(0.3), 32(0.3)]	0.5	0	0.95	0.01	100
TCN Tube	48	64	[64(0.3)] × dilations(1,2,4,8)	0.8	0.001	0.95	0.001	100
GRU Tube	24	64	[64(0.3), 32(0.3)]	0.5	0.005	0.95	0.01	100
Quantile Models	Window Size	Batch Size	Structure	q1	q2	learning rate	epochs	
LSTM Quantile	12	32	[64(0), 32(0)]	0.025	0.975	0.01	100	
TCN Quantile	24	64	[64(0.3)] × dilations(1,2,4,8)	0.025	0.975	0.01	100	
GRU Quantile	48	64	[64(0.3), 32(0.3)]	0.025	0.975	0.01	100	
DeepAR Model	Window Size	Batch Size	Structure	learning rate				
DeepAR	48	16	[64 × 2, 32(ReLU), Gaussian]	1e-3				
MDN Model	Window Size	Batch Size	Structure	learning rate	Components			
MDN	24	16	[64(0)]	0.001	5			

TABLE IV: Hyperparameters for Dataset D data across models

Models	Hyperparameters							
Tube Models	Window Size	Batch Size	Structure	r	δ	q	learning rate	epochs
LSTM Tube	48	16	[64(0.3), 32(0.3)]	0.5	0.001	0.95	0.01	100
TCN Tube	6	32	[64(0.3)] × dilations(1,2,4,8)	0.5	0.018	0.95	0.001	100
GRU Tube	48	64	[64(0.3), 32(0.3)]	0.5	0.01	0.95	0.01	100
Quantile Models	Window Size	Batch Size	Structure	q1	q2	learning rate	epochs	
LSTM Quantile	48	32	[64(0), 32(0)]	0.025	0.975	0.01	100	
TCN Quantile	48	64	[64(0.3)] × dilations(1,2,4,8)	0.025	0.975	0.01	100	
GRU Quantile	48	64	[64(0.3), 32(0.3)]	0.025	0.975	0.01	100	
DeepAR Model	Window Size	Batch Size	Structure	learning rate				
DeepAR	48	32	[64 × 2, 32(ReLU), Gaussian]	1e-3				
MDN Model	Window Size	Batch Size	Structure	learning rate	Components			
MDN	48	64	[64(0)]	0.001	5			